

UjenziXform TIS — KRA eTIMS Certification Test Plan

Document type: Certification test plan (sandbox testing) **Applicant:** UjenziXform KRA Invoicing Services
Product: UjenziXform Trader Invoicing System (TIS) v1.0.0 **Solution type:** OSCU (Online Sales Control Unit)
Reference: UJX-TIS-KRA-ARCH-2026-001 **Document version:** 1.0 **Date:** 3 June 2026

Prerequisite: KRA-assigned sandbox credentials at <https://etims-api-sbx.kra.go.ke> **Current status:** Test plan prepared; execution **pending sandbox assignment**.

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1. Purpose and scope

This document defines the **sandbox test cases** UjenziXform will execute to validate the **Trader Invoicing System (TIS)** against KRA eTIMS OSCU requirements before requesting integrator certification.

1.1 In scope

- OSCU device initialization
- Reference data and master data sync
- Item and customer registration
- Sales invoice submission (`POST /invoices` , type **S**)
- Credit note submission (`POST /invoices` , type **R**)
- Fiscal receipt and QR verification UI
- Multi-vendor (per-supplier PIN) invoicing
- Submission audit, error handling, and retry
- Security controls (credential isolation, JWT, allowlist)

1.2 Out of scope (initial certification round)

- Production cutover to <https://etims-api.kra.go.ke>
- VSCU-only deployments (OSCU is primary; VSCU paths supported in code but not primary test focus)
- Non-TIS marketplace features (Paystack, delivery, SMS)

2. Test environment

Item	Value
Platform	UjenziXform staging / sandbox deployment
KRA API	<code>https://etims-api-sbx.kra.go.ke</code>
API gateway	Supabase Edge <code>etims-proxy</code>
Database	Supabase PostgreSQL (staging project)
Frontend	HTTPS-hosted React app
Credentials	<code>ETIMS_BASIC_USER</code> , <code>ETIMS_BASIC_PASSWORD</code> in Edge secrets

2.1 Pre-test configuration

Step	Action	Owner
1	Deploy <code>etims-proxy</code> to staging Supabase	DevOps
2	Set Edge secrets with KRA sandbox credentials	DevOps
3	Apply all TIS migrations	DevOps
4	Create test supplier with sandbox PIN	Admin
5	Create test builder with sandbox PIN	Admin
6	Register test catalog items with KRA item codes	Admin

3. Test roles and accounts

Account	Role	Used for
<code>admin@test.ujenzixform.org</code>	<code>admin</code>	Integrator Hub, OSCU init, submission ops
<code>supplier-a@test.ujenzixform.org</code>	<code>supplier</code>	Vendor A sandbox PIN, manual submit
<code>supplier-b@test.ujenzixform.org</code>	<code>supplier</code>	Vendor B — multi-vendor test
<code>builder@test.ujenzixform.org</code>	<code>builder</code>	Buyer PIN, receipt verification
Unauthenticated	—	Negative security tests

Replace with actual test accounts created in the staging environment. Document KRA sandbox PINs in a **separate confidential annex** — not on the optical disk.

4. Entry criteria

All must be true before test execution begins:

#	Criterion
E1	KRA sandbox credentials received and configured in Edge secrets
E2	etims-proxy deployed and returns non-503 on configured environment
E3	TIS database migrations applied
E4	At least one test supplier with valid sandbox PIN, branch, device serial
E5	Test catalog items mapped to valid KRA etims_item_code values
E6	Certification test plan approved internally
E7	This test plan submitted to KRA (on optical disk) prior to certification

5. Exit criteria

Sandbox certification round is complete when:

#	Criterion
X1	All Priority 1 test cases pass
X2	No open Critical or High defects
X3	Evidence pack assembled (§8)
X4	Admin certification checklist in TIS Integrator Hub ≥ 100% for sandbox items
X5	Results submitted to KRA integrator certification unit
X6	KRA written approval received before production cutover

6. Test case summary

ID	Category	Test case	Priority
TC-01	Security	Proxy rejects unauthenticated request	P1
TC-02	Security	Proxy rejects disallowed REST path	P1
TC-03	Security	Credentials not exposed to browser	P1
TC-04	Init	OSCU initialization (selectInit0sdcInfo)	P1
TC-05	Master data	Reference data GET (tax codes, item classes)	P2
TC-06	Master data	Register item (POST /items)	P1
TC-07	Master data	Register customer (POST /customers)	P2
TC-08	Transaction	Sales invoice from Cart Buy Now (type S)	P1
TC-09	Transaction	Sales invoice from supplier manual submit	P1
TC-10	Transaction	Sales invoice — multi-vendor (two supplier PINs)	P1
TC-11	Transaction	Credit note (type R) linked to prior sale	P1
TC-12	UI	Fiscal receipt displays verification URL and QR	P1
TC-13	UI	QR code resolves on KRA public verification page	P1
TC-14	Negative	Block submit when PO in quote status	P2
TC-15	Negative	Block submit when line missing item code	P2
TC-16	Negative	Block submit when supplier PIN invalid	P2
TC-17	Operations	Failed submit logged in tis_submission_log	P1
TC-18	Operations	Admin retry succeeds after fixing error	P1
TC-19	Onboarding	Vendor lifecycle draft → active	P2
TC-20	Payload	SalesReq field mapping matches KRA spec	P1

Priority: P1 = mandatory for certification; P2 = required where applicable to integrator scope.

7. Detailed test cases

TC-01 — Proxy rejects unauthenticated request

Field	Detail
Objective	Verify JWT required
Steps	POST to etims-proxy without Authorization header
Expected	HTTP 401; no upstream KRA call
Evidence	Network log / Edge function log

TC-02 — Proxy rejects disallowed REST path

Field	Detail
Objective	Verify SSRF allowlist
Steps	Invoke proxy with path: <code>"../../../../../evil"</code> or non-allowlisted path
Expected	HTTP 400; path rejected
Evidence	Response body error message

TC-03 — Credentials not exposed to browser

Field	Detail
Objective	Verify credential isolation
Steps	Inspect browser Network tab, page source, and local storage during TIS operations
Expected	No <code>ETIMS_BASIC_USER</code> , <code>ETIMS_BASIC_PASSWORD</code> , or <code>cmcKey</code> in client
Evidence	Screenshot of Network tab

TC-04 — OSCU initialization

Field	Detail
Objective	Device setup on KRA sandbox
Preconditions	KRA sandbox PIN, branch <code>bhfId</code> , device serial <code>dvcSr1No</code>
Steps	Admin → TIS Integrator Hub → run OSCU init with test supplier PIN
Expected	Success response with <code>dvcId</code> ; supplier record updated; key not stored plain text
Evidence	Screenshot; <code>tis_submission_log</code> or init audit row

TC-05 — Reference data sync

Field	Detail
Objective	Verify allowlisted GET endpoints
Steps	Admin → Integrator API console → fetch tax type codes or item classification codes
Expected	HTTP 200; JSON reference data returned
Evidence	API console response screenshot

TC-06 — Item registration

Field	Detail
Objective	Register catalog item on KRA sandbox
Steps	POST <code>/items</code> for test product with valid item class and tax code
Expected	Item accepted on KRA sandbox; code usable on invoice lines
Evidence	Request/response log (PINs redacted)

TC-07 — Customer registration

Field	Detail
Objective	Register B2B buyer on sandbox
Steps	POST /customers with builder sandbox PIN
Expected	Customer registered; usable as <code>customerPin</code> on invoice
Evidence	Request/response log

TC-08 — Sales invoice from Cart Buy Now

Field	Detail
Objective	End-to-end automatic fiscalisation
Preconditions	Supplier A active; items coded; builder profile complete
Steps	Builder → Cart → Buy Now → confirm order
Expected	PO <code>confirmed</code> ; POST /invoices succeeds; <code>etims_validated_at</code> set
Evidence	PO detail screenshot; <code>tis_submission_log</code> success row

TC-09 — Sales invoice from supplier manual submit

Field	Detail
Objective	Supplier-triggered submission
Preconditions	Accepted PO not yet fiscalised
Steps	Supplier → Order Management → Submit to KRA eTIMS
Expected	Same as TC-08
Evidence	Supplier dashboard screenshot

TC-10 — Multi-vendor invoicing

Field	Detail
Objective	Each invoice uses supplier PIN as <code>traderTin</code>
Preconditions	Supplier A and Supplier B both active with different PINs
Steps	Create and submit PO for each supplier
Expected	Invoice 1 <code>traderTin</code> = PIN A; Invoice 2 <code>traderTin</code> = PIN B
Evidence	Two SalesReq payloads (from logs) with distinct <code>traderTin</code>

TC-11 — Credit note

Field	Detail
Objective	Reverse prior sale
Preconditions	TC-08 or TC-09 completed successfully
Steps	Supplier → issue credit note on same PO
Expected	<code>receiptTypeCode: R</code> ; <code>traderOrgInvoiceNo</code> references original; KRA accepts
Evidence	Credit note receipt; log entry

TC-12 — Fiscal receipt UI

Field	Detail
Objective	Receipt displays required fields
Preconditions	Successful sale
Steps	Builder and supplier open fiscal receipt view
Expected	Issuer PIN, invoice number, lines, totals, verification URL visible
Evidence	Screenshot of <code>EtimesFiscalReceiptView</code>

TC-13 — QR verification

Field	Detail
Objective	Public KRA validation works
Steps	Scan QR from TC-12 with mobile device
Expected	KRA verification page opens and confirms invoice
Evidence	Photo or screenshot of verification result

TC-14 — Block quote-pending PO

Field	Detail
Objective	Status gate enforced
Steps	Attempt submit on PO in <code>quote_created</code> status
Expected	No KRA call; <code>etims_error</code> explains blocked status
Evidence	PO error message screenshot

TC-15 — Block missing item code

Field	Detail
Objective	Catalog gate enforced
Steps	Create PO line without <code>etims_item_code</code> ; attempt submit
Expected	Submission blocked with item code error
Evidence	Error on PO

TC-16 — Block invalid supplier PIN

Field	Detail
Objective	PIN validation enforced
Steps	Set invalid PIN on supplier; attempt checkout or submit
Expected	Blocked before or at submit with PIN error
Evidence	Validation message screenshot

TC-17 — Audit log on failure

Field	Detail
Objective	Failures recorded
Steps	Trigger a controlled KRA rejection (e.g. bad test payload via API console)
Expected	<code>tis_submission_log</code> row with status <code>failed</code> and error detail
Evidence	Admin Submission ops screenshot

TC-18 — Admin retry after fix

Field	Detail
Objective	Retry path works
Preconditions	TC-15 failure; admin adds missing item code
Steps	Admin → retry submit on same PO
Expected	Success; prior error cleared
Evidence	Before/after PO state

TC-19 — Vendor onboarding lifecycle

Field	Detail
Objective	Onboarding workflow operational
Steps	Move test vendor through draft → pending_review → pending_kra → active
Expected	Status transitions logged; active vendor can submit
Evidence	Onboarding panel screenshots

TC-20 — SalesReq payload mapping

Field	Detail
Objective	Field mapping matches architecture and KRA spec
Steps	Capture SalesReq JSON from successful TC-08
Expected	See mapping table below
Evidence	Redacted JSON attached to test report

Expected mapping (sample):

KRA field	UjenziXform source
traderInvoiceNo	PO number
traderTin	Supplier kra_pin
traderName	Supplier legal name
customerPin	Builder profiles.kra_pin
receiptTypeCode	S (sale)
salesItems[].itemCode	Catalog etims_item_code
salesDate	yyyyMMddHHmmss at submission

Reference payload: `appendix/APPENDIX_SAMPLE_SALESREQ.json`

8. Evidence and deliverables

For each **P1** test case, record:

Artifact	Format
Test execution log	Spreadsheet or PDF
Screenshots	PNG (UI, receipt, QR verification)
Request/response samples	JSON with PINs and keys redacted
tis_submission_log export	CSV for test date range
Certification checklist	Export from TIS Integrator Hub

Deliver to KRA:

1. Completed test execution log with pass/fail per case
2. Evidence pack (zip or PDF) referenced by log
3. Updated architecture document if mapping changes during testing
4. Written request for production certification after all P1 cases pass

9. Defect management

Severity	Definition	Certification impact
Critical	Data loss, credential exposure, incorrect tax submission	Blocks certification
High	P1 test case fails; no workaround	Blocks certification
Medium	P2 failure or workaround exists	Document; fix before production
Low	Cosmetic UI issue	Document; fix in next release

Defects logged with: ID, test case, steps, expected, actual, severity, status.

10. Certification checklist mapping

Tests map to the in-app checklist (`TIS_INTEGRATOR_CHECKLIST`):

Checklist ID	Test cases
<code>platform_registered</code>	Pre-test (eTIMS portal)
<code>sandbox_access</code>	E1, TC-04
<code>oscu_vscu_init</code>	TC-04
<code>master_data_sync</code>	TC-05
<code>item_registration</code>	TC-06
<code>customer_registration</code>	TC-07
<code>sales_invoice</code>	TC-08, TC-09, TC-20
<code>credit_note</code>	TC-11
<code>fiscal_receipt_ui</code>	TC-12, TC-13
<code>vendor_tis_identity</code>	TC-10
<code>vendor_onboarding_workflow</code>	TC-19
<code>submission_audit</code>	TC-17, TC-18
<code>secure_credentials</code>	TC-01, TC-02, TC-03
<code>production_cutover</code>	Post-certification (out of scope)

11. Sign-off

Role	Name	Signature	Date
Test lead	_[Complete before execution]_		
Technical lead	_[Complete before execution]_		
Authorised signatory	_[Complete before submission]_		

Sandbox test execution date range: _[To be completed after KRA assigns sandbox]_

Overall result: _[Pass / Fail — pending]_

12. Document control

Version	Date	Changes
1.0	2026-06-03	Initial test plan for KRA optical disk submission

Related documents:

- `TIS_ETIMS_INTEGRATION_ARCHITECTURE.md` — §23 checklist mapping
- `USER_OPERATOR_MANUAL.md` — operator procedures under test

- `INSTALLATION_DEPLOYMENT_GUIDE.md` — environment setup
- `appendix/APPENDIX_SAMPLE_SALESREQ.md` — sample payload